AUTONOMOUS FIRE FIGHTING ROBOT

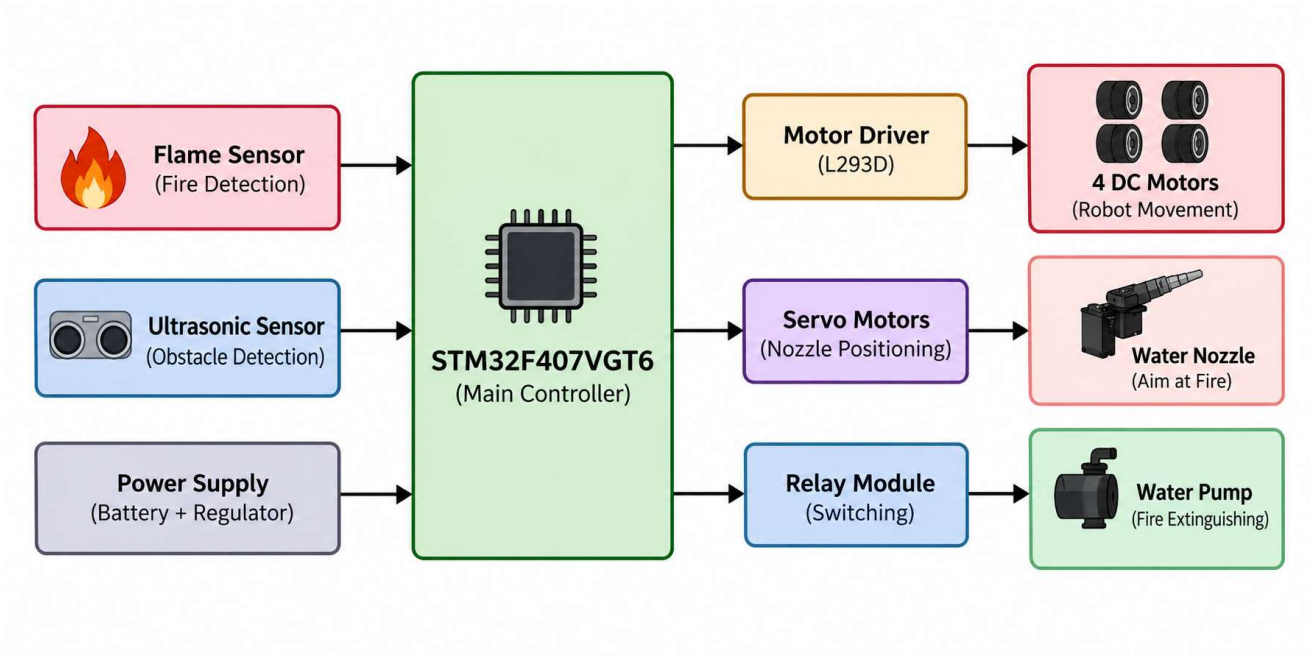
Technical Overview | STM32F407VGT6 • Embedded Control • Fire Detection and Control

Project at a glance

Purpose: detect fire using flame sensors and coordinate robot movement, nozzle positioning and water-pump activation through an STM32-based control system.

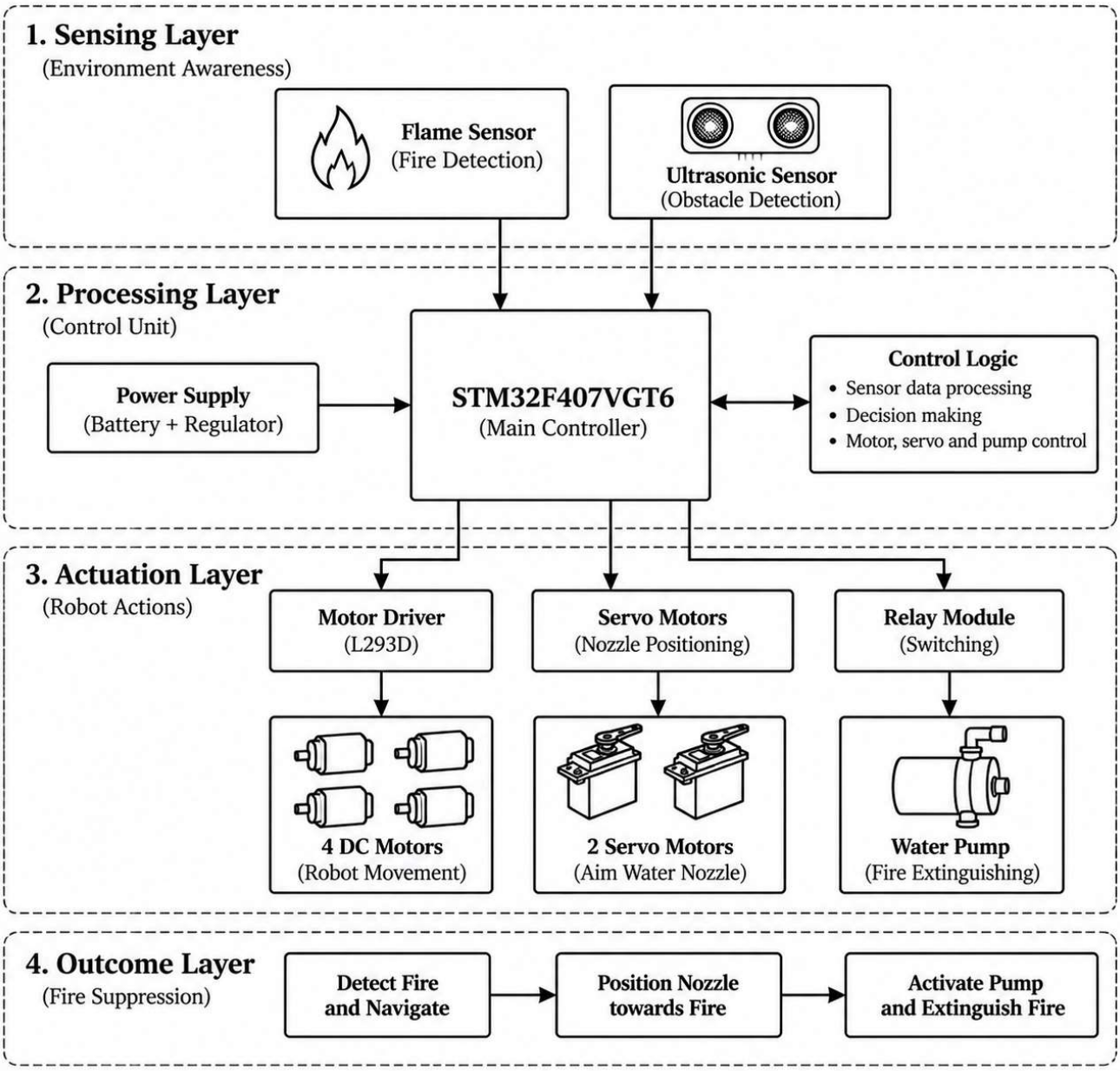
Contents: system block diagram, layered architecture, circuit diagram, operational flowchart and three reference tables covering component interfaces, IoT features and status indicators.

1. System Block Diagram



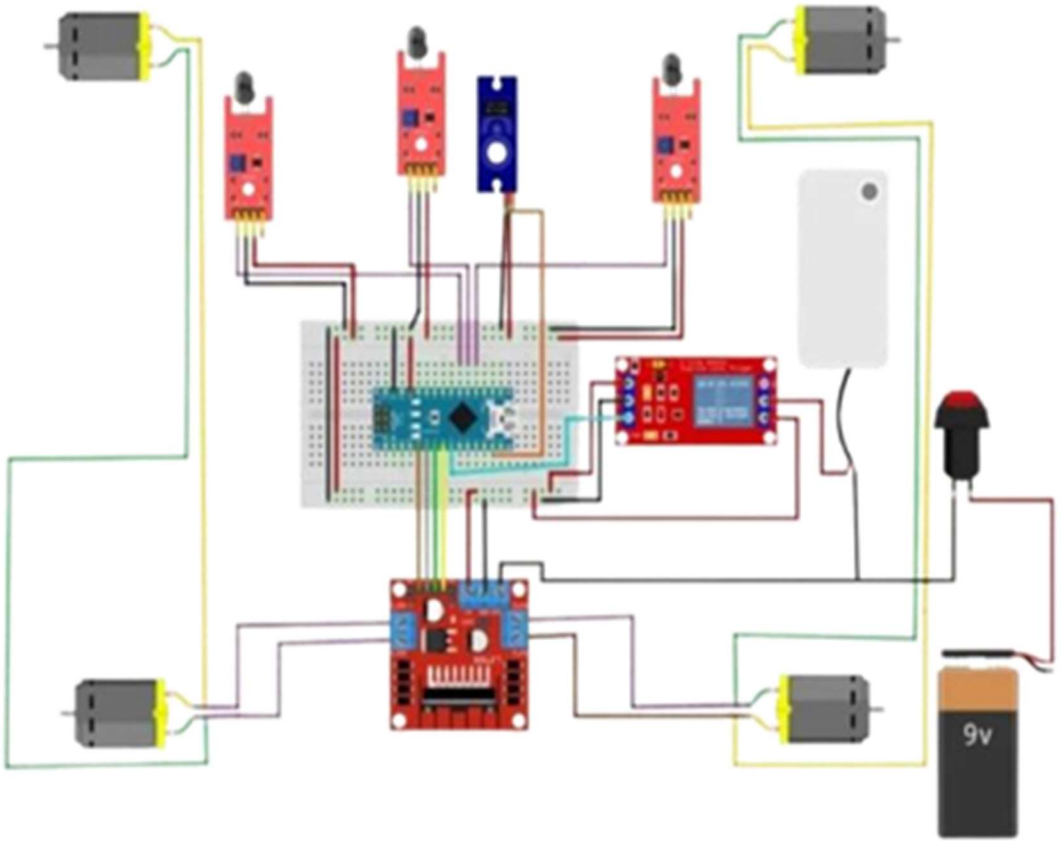
The STM32F407VGT6 receives flame-sensor inputs and controls the motor driver, servo motors and relay-operated water pump.

2. System Architecture



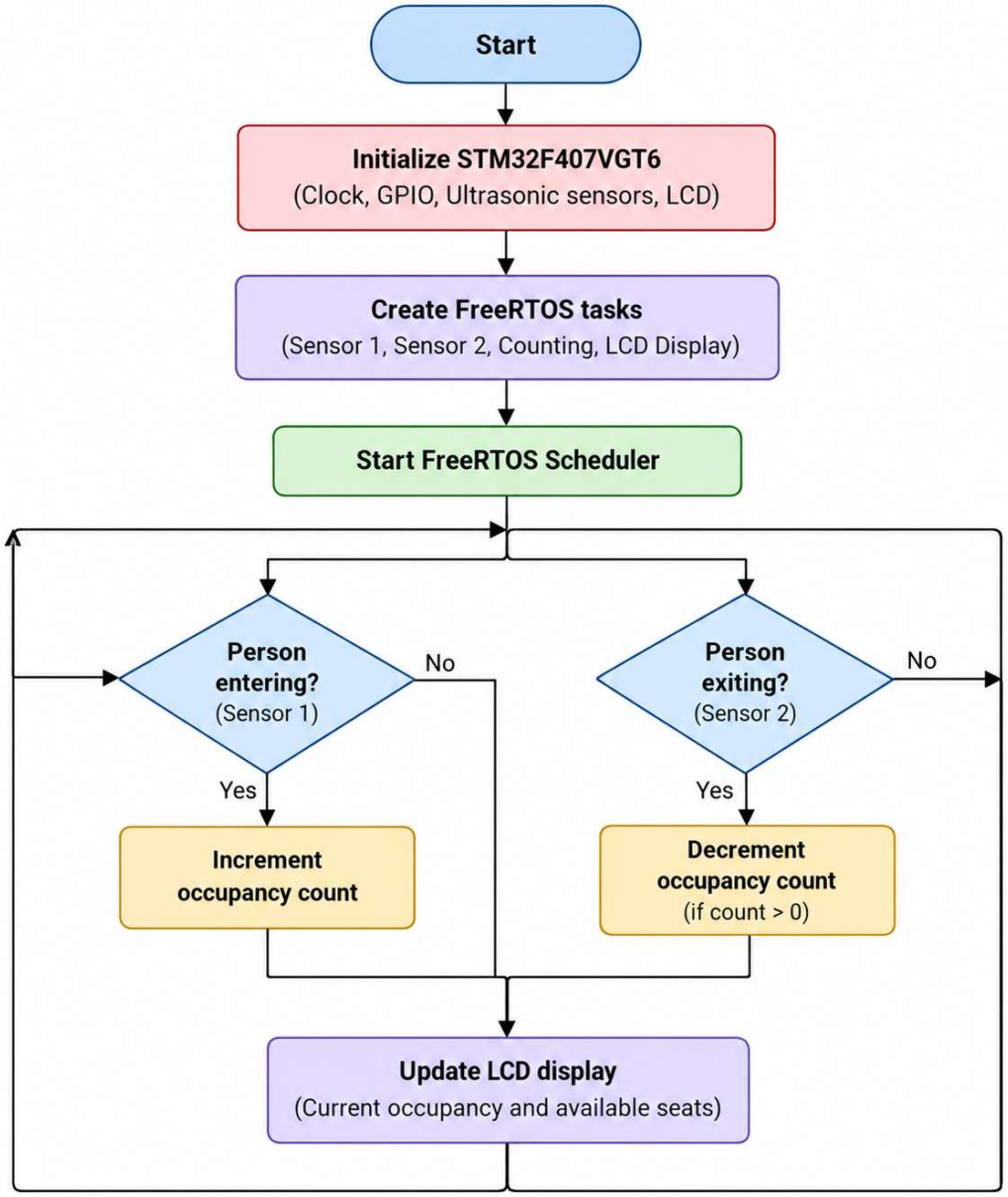
The architecture separates sensing, STM32 processing, actuation and system outcome functions.

3. Hardware Circuit Diagram



Circuit reference supplied for the project prototype. Confirm the exact controller board, sensor connections, power arrangement and driver wiring against the assembled hardware before using it as a wiring guide.

4. Operational Flowchart



The flowchart summarizes initialization, flame monitoring, movement/scan operation and the fire-response sequence.

5. Component Interface and Control

COMPONENT	INTERFACE TYPE	CONTROLLED BY	FUNCTION
Flame Sensor	Analog	ADC on STM32	Detects flame intensity and location
Motor Driver (L293D)	PWM / GPIO	STM32	Controls DC motors
Servo Motors	PWM	STM32	Aims nozzle and sensor direction
Relay Module	GPIO	STM32	Switches water pump ON/OFF

6. IoT and Monitoring Features

FEATURE	DESCRIPTION	FUNCTIONALITY
Cloud Database	Stores sensor data and status logs	Real-time and historical analysis
Web Dashboard	Displays robot data visually	Shows fire detection events and motor activity
Remote Control	Allows manual override	Control pump or standby remotely
IoT Connectivity	Wi-Fi / ESP32 communication	Sends STM32 data to the cloud

7. Status Indicators and Display

COMPONENT	FUNCTION	INDICATOR COLOR / DISPLAY
Power LED	Indicates system power ON	Red/Green
Fire Detection LED	Lights up when flame is detected	Red
Pump LED	Shows pump activation	Blue
Serial Monitor / Display	Displays flame intensity, motor status	Text output via UART